1. Code the variables definition and the process for each of the following:

a) A floating point named price initialized to 9.90.

b) An integer named quantity.

c) A constant integer named DISC initialized to 10.

d) A floating point name totalprice. Get the quantity purchased from the user and then, calculate the totalprice using the following formula:

totalprice = (price*quantity) * (100-DISC)/100.00;

exercise1 (Global Scope)

```cpp
#include <iostream>
using namespace std;

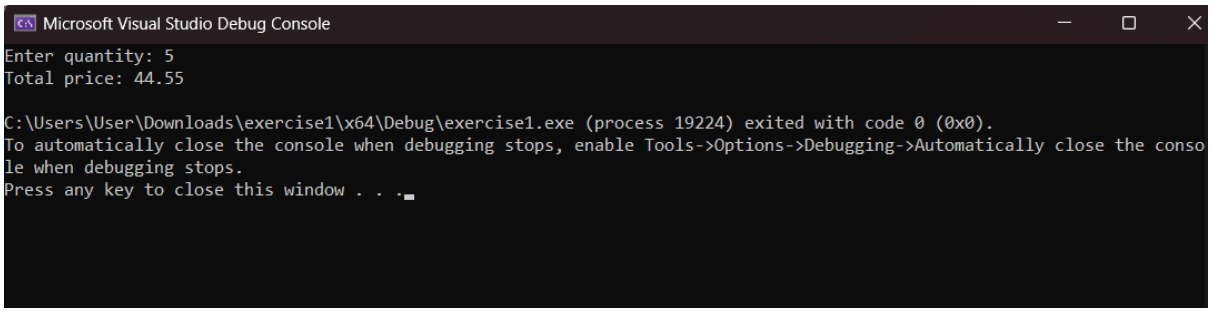
int main()
{
    double price = 9.90;
    int quantity;
    const int DISC = 10;
    double totalprice;

    cout << "Enter quantity: ";
    cin >> quantity;

    totalprice = (price * quantity) * (100 - DISC) / 100.00;

    cout << "Total price: " << totalprice << endl;

    return 0;
}
```



2. Write a program that prompts the user to enter the total time in seconds and displays the hours, minutes and seconds for the total time. Declare two constant integers named SCND_TO_HOUR initialized to 3600 and SCND_TO_MINUTE initialized to 60. For example, 5000 seconds are equal to 1 hour, 23 minutes and 20 seconds. Here is a sample run:

Input the total time in seconds : 5000 5000 seconds are equal to 1 hours 23 minutes and 20 seconds

exercise2 (Global Scope)

```cpp
#include <iostream>
using namespace std;

int main()
{
    const int SCND_TO_HOUR = 3600;
    const int SCND_TO_MINUTE = 60;

    int totalSeconds, hours, minutes, seconds, remaining;

    cout << "Input the total time in seconds : ";
    cin >> totalSeconds;

    hours = totalSeconds / SCND_TO_HOUR;
    remaining = totalSeconds % SCND_TO_HOUR;
    minutes = remaining / SCND_TO_MINUTE;
    seconds = remaining % SCND_TO_MINUTE;

    cout << totalSeconds << " seconds are equal to "
        << hours << " hours, " << minutes << " minutes and "
        << seconds << " seconds." << endl;

    return 0;
}
```

Microsoft Visual Studio Debug Console

```
Input the total time in seconds : 5000
5000 seconds are equal to 1 hours, 23 minutes and 20 seconds.

C:\Users\User\Downloads\exercise2\x64\Debug\exercise2.exe (process 22468) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

3. Write a program that reads in loan amount, annual interest rate and number of years for the loan and display monthly payment the customer should pay for the loan. Here is a sample run:

Input loan amount (RM): 30000 Input annual interest rate (%) : 4.2 Input number of years : 10 Monthly payment for this loan is (RM): 355

```cpp
#include <iostream>
using namespace std;

int main()
{
    const int MONTHS_IN_YEAR = 12;

    double loanAmount, annualRate, totalPayment, monthlyPayment;
    int years;

    cout << "Input loan amount (RM): ";
    cin >> loanAmount;

    cout << "Input annual interest rate (%) : ";
    cin >> annualRate;

    cout << "Input number of years : ";
    cin >> years;

    totalPayment = loanAmount + (loanAmount * annualRate / 100 * years);
    monthlyPayment = totalPayment / (years * MONTHS_IN_YEAR);

    cout << "Monthly payment for this loan is (RM): " << monthlyPayment << endl;

    return 0;
}
```

```
Input loan amount (RM): 30000
Input annual interest rate (%) : 4.2
Input number of years : 10
Monthly payment for this loan is (RM): 355

C:\Users\User\Downloads\exercise3\x64\Debug\exercise3.exe (process 13396) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

4.In physics, an object that is in motion is said to have kinetic energy. The following formula can be used to determine a moving object's kinetic energy:

KE = (1/2 mv2)

Where

KE = the kinetic energy

m = the object's mass in kilograms

v = the object's velocity in meters per second.

Write a program to calculate the amount of kinetic energy that the object has when given the object's mass (in kilograms) and velocity (in meters per second). Here is a sample run:

```
Input the object's mass (kg):10.0
Input the object's velocity (m/s): 5
The kinetic energy (Joules): 125
```

exercise4 (Global Scope)

```cpp
#include <iostream>
using namespace std;

int main()
{
    double mass, velocity, kineticEnergy;

    cout << "Input the object's mass (kg): ";
    cin >> mass;

    cout << "Input the object's velocity (m/s): ";
    cin >> velocity;

    kineticEnergy = 0.5 * mass * velocity * velocity;

    cout << "The kinetic energy (Joules): " << kineticEnergy << endl;

    return 0;
}
```

Microsoft Visual Studio Debug Console

```
Input the object's mass (kg): 10
Input the object's velocity (m/s): 5
The kinetic energy (Joules): 125

C:\Users\User\Downloads\exercise4\x64\Debug\exercise4.exe (process 2848) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```